Girder Alignment Report

As-left record · generated 2026-09-04 10:24 UTC

Identification

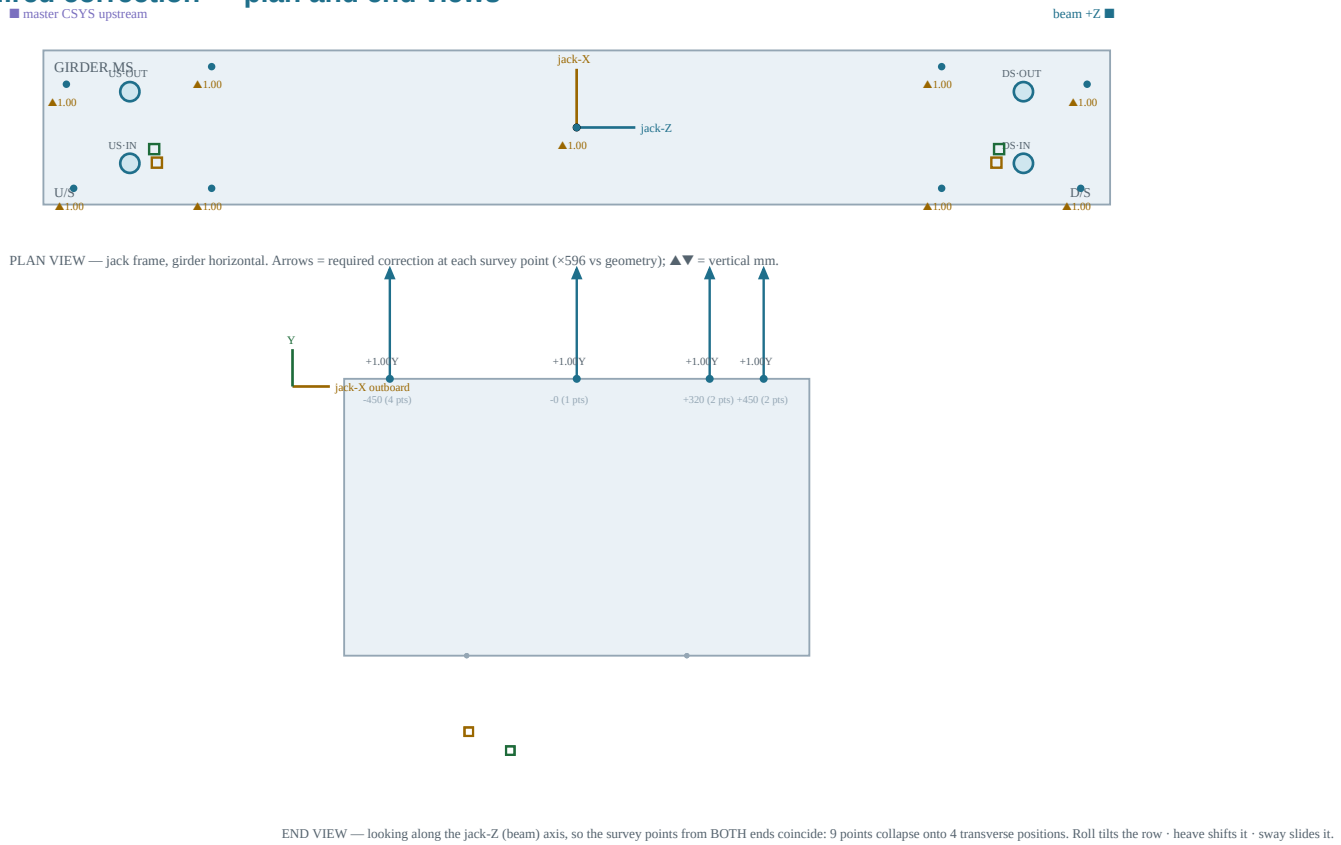
Girder serial	DLS0011116	Operator	aaa
Girder type	MS	Session id	036e9522d245
Started	2026-09-04T10:23:17+00:00	Status	complete
Geometry revision	2026-09-02-r2	Software	2.0.0
Bay	1	Temperature	21.5 °C
Iterations	2	Moves	4

Pose — initial vs as-left

Axis	Unit	Tolerance	Initial error	As-left	% of tol	Status
Roll	mrad	±0.05	+0.000	+0.000	0%	OK
Sway	mm	±0.1	-0.000	-0.000	0%	OK
Heave	mm	±0.1	-1.000	+0.000	0%	OK
Yaw	mrad	±0.15	+0.000	+0.000	0%	OK
Pitch	mrad	±0.15	-0.000	+0.000	0%	OK
Surge	mm	±0.2	-0.000	-0.000	0%	OK

Errors are the girder pose relative to target, in the jack frame. Angles in mrad, translations in mm. Priority order: roll, sway, heave, yaw, pitch, surge.

Required correction — plan and end views



Arrows are the correction at each survey point (measured → nominal). Both views are in the jack frame: the plan view puts the girder horizontal, the end view looks downstream along the beam. In-plane arrows are exaggerated to be visible; in the plan view the vertical component is labelled numerically in mm.

Survey history

#	Time (UTC)	File	Pts	RMS mm	Max mm	Roll	Sway	Heave	Yaw	Pitch	Surge
1	2026-09-04 10:23:22	MS_REF_PTS_-1mm_VE	9	0.0001	0.0001	+0.000	-0.000	-1.000	+0.000	-0.000	-0.000
2	2026-09-04 10:24:52	MS_REF_PTS.csv	9	0.0001	0.0001	+0.000	-0.000	+0.000	+0.000	+0.000	-0.000

Moves — target vs achieved

Iter	Step	Encoder	Target mm	Achieved mm	Error mm	WARP	STRETCH	Gate
1	combined_first	V_US_IN	-0.951	-0.949	+0.0014	-0.0002	+0.0003	pass
		V_US_OUT	-0.951	-0.950	+0.0012			
1	combined_second	V_DS_IN	-1.000	-0.997	+0.0025	-0.0001	-0.0003	pass
		V_DS_OUT	-1.000	-0.997	+0.0030			
1	sway	SWAY_US	+0.000	-0.000	-0.0004	-0.0002	+0.0000	pass
		SWAY_DS	-0.000	-0.001	-0.0004			
1	surge	SURGE_US	+0.000	-0.000	-0.0001	+0.0001	-0.0003	pass
		SURGE_DS	-0.000	+0.000	+0.0004			

Achieved values are encoder readings relative to the datum captured at the start of each move. Only the last pair moved lands on final targets; earlier pairs stop at a compensated intermediate value (cross-shift ≈ 0.0492 mm per mm of far-pair travel).

Model validation — predicted vs achieved

Step	Axis	Predicted	Actual	Residual
surge	roll	-0.001	-0.000	+0.001
	sway	+0.000	-0.000	-0.000
	heave	+1.000	+1.000	+0.000
	yaw	+0.000	-0.000	-0.000
	pitch	-0.000	+0.000	+0.000
	surge	-0.000	-0.000	+0.000

Achieved encoder deltas pushed through the Jacobian and compared with the next survey. Large residuals indicate the geometric model or the bearing pivot assumption needs review.

Uploaded survey data (last survey)

Reference	X	Y	Z	Residual mm
APNT26	-403.092	-450.000	1365.051	0.0001
APNT27	368.788	-450.000	1365.881	0.0001
APNT28	-475.378	-450.000	2383.066	0.0000
APNT29	422.361	-450.000	2446.812	0.0000
APNT30	-217.747	-450.000	5108.158	0.0000
APNT31	-857.855	-450.000	7769.504	0.0001
APNT32	39.885	-450.000	7833.250	0.0001
APNT33	-166.001	-450.000	8897.345	0.0001
APNT34	-930.731	-450.000	8795.830	0.0001

Points are matched to reference fiducials by proximity (gate 10 mm), not by name — names repeat between girder types.

Sign-off

Aligned by	aaa	Signature	
Checked by		Signature	
Date		Notes	